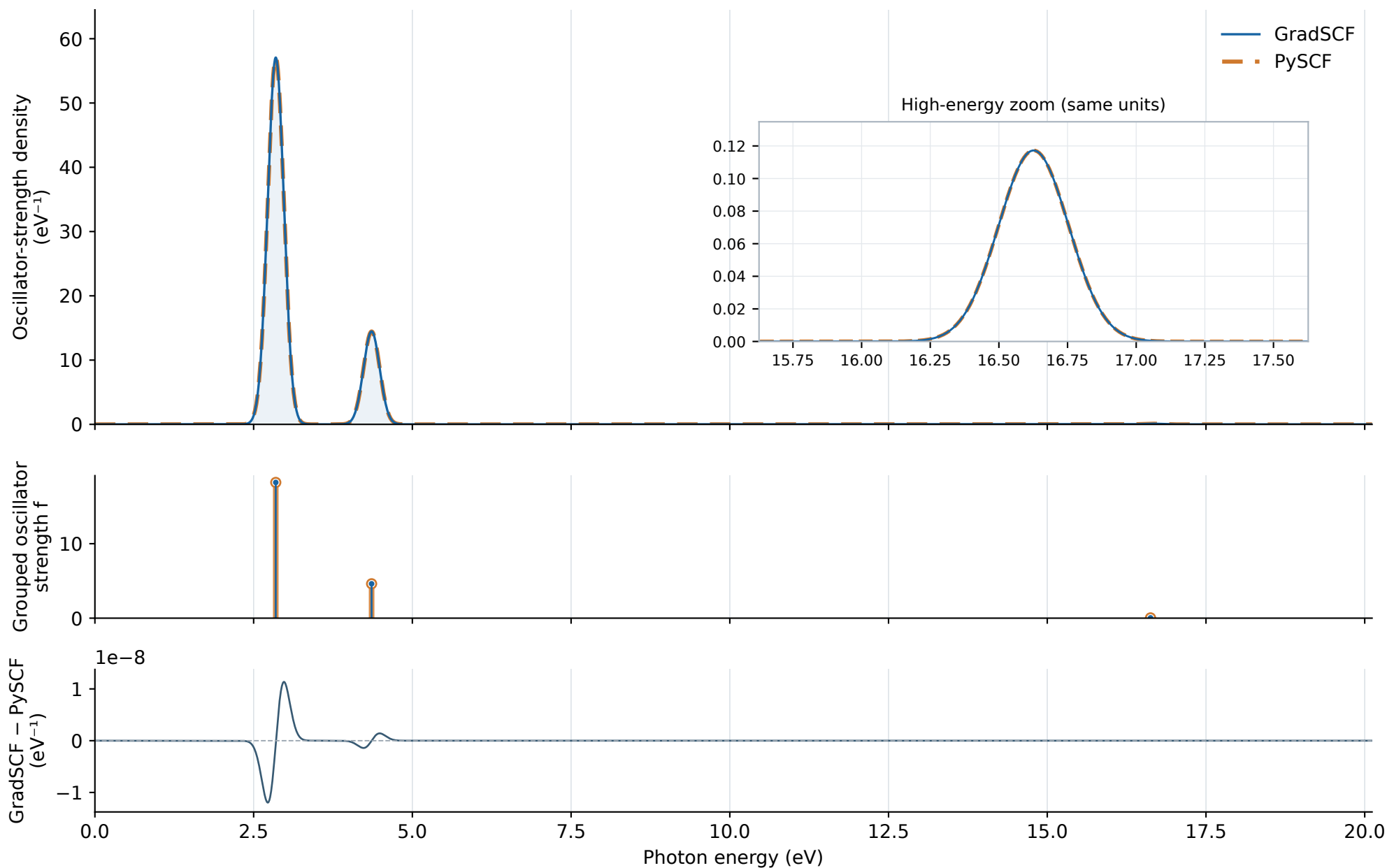


Periodic Si: velocity-gauge TDDFT spectrum

Γ point • PBE / GTH-SZV / GTH-PBE • FFT 41^3 • 16 singlet states

Gaussian FWHM = 0.30 eV | Peak-normalized max difference = 2.10×10^{-10}



Unit-area broadening; absolute oscillator strengths retained. Both include the nonlocal GTH velocity correction.

Γ -point per-cell spectrum with degenerate-state grouping. Bulk k integration and macroscopic absorption coefficients are not computed.